Lóránt Nagy, PhD

lorant-nagy.github.io

Contact Information

E-mail:	lorantnagy@renyi.hu
Mailing:	Alfréd Rényi Institute of Mathematics Reáltanoda utca 13-15, H-1053 Budapest, Hungary

Education and Employment

2015 - 2018	MSc, Mathematics	Budapest University of Technology and Economics
2017 - 2018	Internship	Morgan Stanley & Co LLC
2018 - 2022	PhD, Mathematics	Central European University
2024 .	Research Assistant	Artificial Intelligence National Laboratory
2019 -	Research Assistant	Alfred Renyi Institute of Mathematics
2023 -	Research Assistant	Eötvös Lóránd University – AI Research Group

Publications and Research

Rásonyi, M, Nagy, L, *Optimal long-term investment in illiquid markets when prices have negative memory*, Electronic Communications in Probability. 2021; 26

Guasoni, P, Nagy, L, Rásonyi, M. *Young, timid, and risk takers* Mathematical Finance. 2021; 31: 1332–1356.

Rasonyi, M, Nagy, L, *On arbitrage with linear and superlinear friction* In preparation.

Boros D, Csanády B, Ivkovic I, Nagy L, Lukács A, Márkus L. *Deep learning the Hurst parameter of linear fractional processes and assessing its reliability* Qual Reliab Engng Int. 2024; 40: 4228–4246.

Csanády, B, Nagy, L, Boros, D, Ivkovic, I, Kovács, D, Tóth-Lakits, D, Márkus, L, Lukács, A. *Parameter Estimation of Long Memory Stochastic Processes with Deep Neural Networks* ECAI 2024. 2024; 392

Rásonyi, M, Nagy, L *Optimal investment with transient price impact and stochastic spread* arxiv at 2511.12093

Collaborations and Technical Skills

Morgan Stanley (interest rate modeling), **PyTorch**, *Wolfram Mathematica*, **ALTEO** (forecasting intraday prices), **Docker**, **Sysler Kft** (disease progression modeling), Rust, C++, **Python**

Seminar and Conference Talks

2020; *How to invest in mean-reverting markets?*, **Alfréd Rényi Institute of Mathematics**, Probability Seminar

2020; *Investment in mean-reverting markets*, **Central European University**, PhD Seminar

2021; *On sub-Gaussian estimators of the mean*, **Alfréd Rényi Institute of Mathematics**, Probability Seminar

2022; *Young, Timid, and Risk Takers*, **University of Michigan**, Financial Mathematics Seminar

2023; *Random Walks in Random Environments*, **Alfréd Rényi Institute of Mathematics**, Probability Seminar

2023; *Risk averse trading*, **Eötvös Lóránd University**, ELTE-Rényi Mi Workshop

2024; *Optimal execution in superlinearly mean-reverting environments*, **Bolyai Intézet**, AMK

2025; *Non-convex optimal investment with transient price impact*, **Technische Universität Wien**, VCMF2025

2026; *On the utility problem in a market where price impact is transient*, Bologna, *XIII* Bachelier Congress

Miscellaneous

2025; Intermediate-level **Qualification in Copyright Law**, Hungarian Intellectual Property Office

2022; Award for Advanced Doctoral Students

2019; Teaching Fundamentals of Stochastic Analysis course at the Central European University

2016; Second Place Award, TDK competition, ”Examples in the theory of continuous trading with friction”